

How Trade Policy Uncertainty Regimes Reshape Stock Sector Network Structure

Data & AI in Economics — Group H

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Motivation & Economic Relevance

The standard treatment of Trade Policy Uncertainty (TPU) focuses on how it shifts return distributions or elevates volatility for *individual* assets. We argue this misses the key structural question: **does uncertainty change how risk propagates across the entire equity system?**

Research Question

How does TPU alter the correlation structure and directional interdependencies across US stock market sectors, and which sectors emerge as central hubs or isolated absorbers during high-uncertainty regimes — as identified through the 2025 tariff events?

Why it matters. The 2025 tariff cycle against China, Canada and Mexico was the sharpest trade escalation in decades. Understanding whether it rewired sectoral interdependencies — not just shifted average returns — has direct implications for systemic risk monitoring and regime-aware portfolio construction.

Data. Daily observations 2018–2026 covering: 11 US Select Sector SPDR ETFs (XLK, XLI, XLY, ...), the Baker–Bloom–Davis TPU index, VIX and the 10y–2y interest rate spread from FRED. A one-day publication lag is enforced on all macro variables to prevent look-ahead bias.

Unsupervised

Hidden Markov Model on $\log(\text{TPU})$
 $k = 3$ states, BIC-selected
TVTP extension:
transition probs conditioned on VIX
K-Means & Hierarchical clustering as benchmarks

Causal Inference

DAG (DoWhy): VIX and rate spread as confounders
Backdoor adjustment by partialling out to obtain “purged” returns
Granger causality networks per regime (BH-FDR, $q = 0.05$)
Johansen test & VECM on log-prices

Supervised Learning

Target: directional weekly sector return (binary)
Features: lagged returns, realized vol, HMM regime label, network centrality
Models: Logistic, Ridge/Lasso, Neural Network
Eval: F1, ROC-AUC, expanding-window CV

Results I: Regime Detection

HMM identifies three economically distinct TPU states. The BIC-selected 3-state univariate HMM on $\log(\text{TPU})$ produces well-separated emission means and highly persistent states. The High-TPU regime (mean ≈ 1200) covers 29.3% of the sample and aligns precisely with the two major trade escalation windows.

	Low	Mid	High
Mean TPU	197	402	1200
Long-run prob.	28.1%	29.7%	42.2%
Expected duration	25 d	27 d	453 d
Mean VIX	15.3	20.7	22.9
VIX std. dev.	3.4	4.1	9.8

TVTP extension. Conditioning transition probabilities on VIX significantly improves model fit (LR = 11.00, $p = 0.0041$). **VIX governs entry into the High-TPU regime** (entry prob. peaks near 0.85–0.9 during 2020 and 2025), but has almost no effect on exit — once started, the regime persists on its own time scale.

Place HMM regime plot here.

Results II

Causal Inference

TPU rewires sector interactions

Network densifies $11\times$. BH-FDR ($q = 0.05$) on 110 Granger tests per regime: 2 edges in Non-High-TPU (density 0.018) vs 22 in High-TPU (density 0.200).

Hub-and-spoke topology. XLP and XLU absorb (in-degree 9 and 8), joint PageRank $\approx 46\%$. XLY transmits (out-degree 4).

Equilibrium tighter, leadership rotates. Johansen rank = 1 both regimes (trace 309.87 vs 292.08). VECM leaders shift from growth (XLY, XLK, XLC) to real-economy (XLRE, XLI, XLE).

Two horizons. Short-term absorbers $\neq$ long-term leaders.

Supervised Learning

Regimes carry predictive signal

Leak-free validation. Purged, embargoed CV; naive baseline at 0.500 ROC-AUC as expected.

Linear $>$ neural net. Lasso/Ridge (AUC ≈ 0.545) edge out the NN (≈ 0.537); TVTP features lift AUC $0.536 \rightarrow 0.545$.

Tariff shock (Phase 5). Event-window networks rewire significantly when tariffs take effect (Mar 4, $p \approx 0.003$); marginal at Apr 2 ($p \approx 0.08$).

Cross-check. Post-shock network resembles the High-TPU regime ($\text{sim}_{\text{High}} > \text{sim}_{\text{Low}}$).

Limitations & Synthesis

Unsupervised

- TVTP vs. baseline is not just a fit question (LR favors TVTP, BIC the simpler model): it reflects whether regime-switching is autonomous or VIX-triggered
- Modeling $\log(\text{TPU})$ as discrete regimes (vs. a continuous trend) is a representational choice the borderline ADF result ($p = 0.074$) does not resolve on its own

Causal Inference

- Granger captures only linear dependencies; lag fixed at $p = 5$
- Backdoor set limited to VIX and 10y–2y spread; other macro confounders not modeled
- Regime labels from TVTP-HMM treated as ground truth, no propagation of classification uncertainty

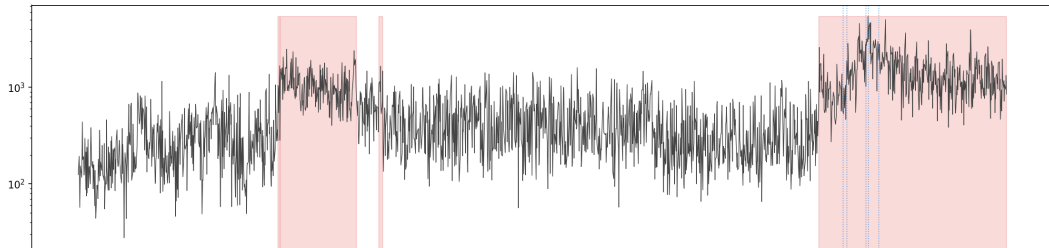
Supervised Learning

- Best AUC ≈ 0.545 ; noisy weekly returns inherently bound absolute predictive performance.
- High sector correlation reduces effective sample size, despite strictly purged CV folds.
- Short ($\pm 30\text{d}$) event windows; Apr 2 overlaps macro shocks, so rewiring isn't tariff-only.

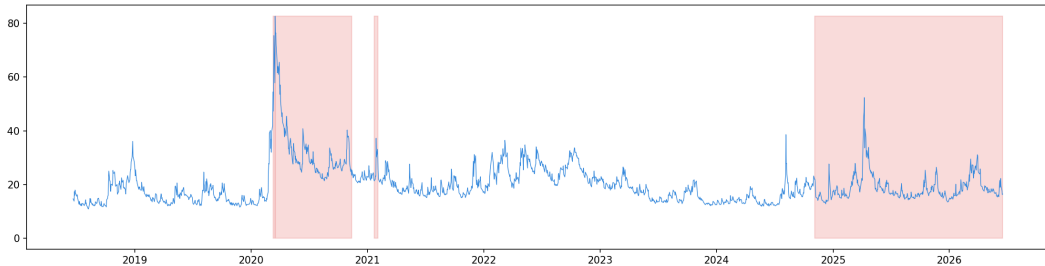
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HMM (TVTP) K=2

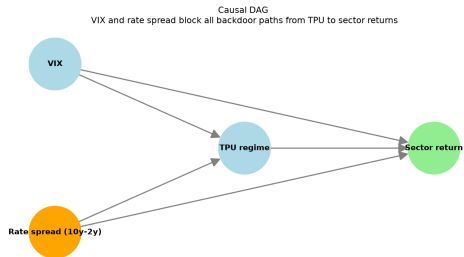
TVTP-HMM — Time-Varying Transition Probabilities (Filardo 1994)



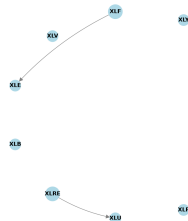
E — VIX



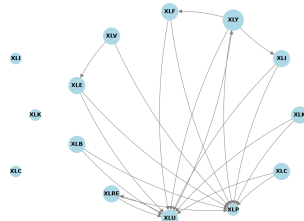
DAG and Network



granger network — Non-High-TPU
2 edges, density=0.018

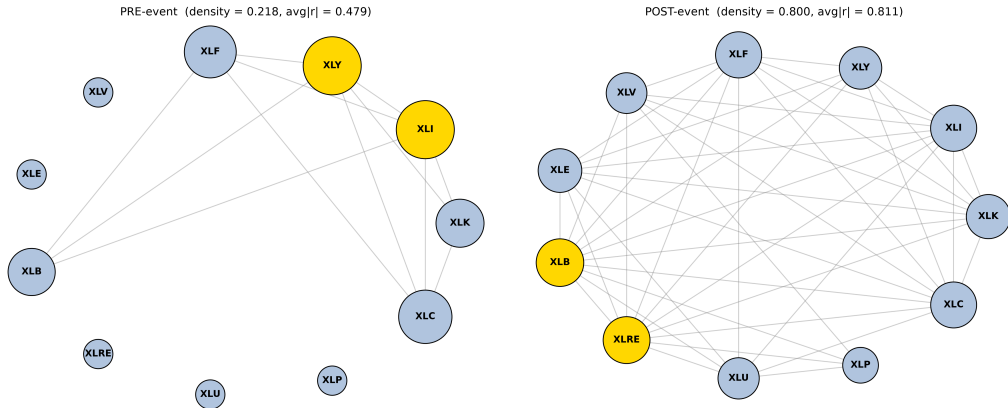


granger network — High-TPU
22 edges, density=0.200



Tariff Event Study — Sector Network Rewiring

**Sector Correlation Network Rewiring around
Apr 2 2025: Liberation Day (window = 30 trading days, $|r| > 0.7$)**



Pre- vs. post-event correlation networks (node size = PageRank centrality). Density jumps from 0.22 to 0.80; the rewiring is statistically strongest when the tariffs take effect (Mar 4, permutation $p \approx 0.003$).